Helena (Nellie) Kolon

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Summary

Scientist and technical leader who brings deep lab and engineering expertise to AI-driven health products. Five-plus years of assay development and high-throughput experimentation across pharma (Eli Lilly, Takeda), academia (Georgia Tech), and a health AI startup, where I lead R&D, manage a team, and partner with ML engineers to turn biological data into models.

Education

Georgia Institute of Technology – M.S. Biomedical Engineering — Aug '24 – Dec '25

Northeastern University – B.S. Biochemistry, Magna Cum Laude — Sep '17 – May '21

Skills

AI and data: AI-assisted experimental design and optimization, human-in-the-loop workflows, data quality and ground-truth generation for ML models, domain expertise for model design, Design of Experiments (DoE)

Leadership: hiring, training, and managing research interns, setting technical roadmaps and priorities, cross-functional collaboration with ML and engineering, external partnerships, scientific presentations

Lab and instrumentation: CE-LIF, microfluidics, HPLC/MS, high-throughput screening and liquid handling, tangential flow filtration, ELISA, in vitro and in vivo assays, bioconjugation, nanoparticle formulation and scale-up

Experience

Deleon Technologies – Atlanta, GA — Sep '25 – Present

Founding Engineer

- Built the chemical analysis system for our miniaturized CE-LIF device, from idea to working instrument.
- Developed the assay from scratch: 10+ amino acids in a 2-minute run, within 3% of a commercial instrument.
- Hired, trained, and manage two Georgia Tech research interns, who ran the device independently within a month.
- Domain expert for the ML team, translating metabolic pathways and kidney physiology into model features.
- Moved the lab from trial-and-error to AI-guided experiment design, adding context the model can't see.

Georgia Institute of Technology: Ackun-Farmmer Lab – Atlanta, GA — Aug '24 – May '25

Graduate Student Researcher

- Cut nanoparticle synthesis from 10 hours to 30 min by updating long-standing protocols with recent literature.
- Designed peptide-targeted nanoparticles to retrain immune cells in multiple sclerosis.
- Co-wrote a Biomaterials Science review, including the sections on lipid-based materials.
- Mentored an undergraduate in synthesis and in vitro techniques and helped him earn a research scholarship.

Eli Lilly and Company: Genetic Medicine – Cambridge, MA — May '21 – May '24

Senior Research Associate

- Helped build an automated LNP platform with automation engineers, from 5 formulations a day to 100+ a week.
- Generated data for an ML model of lipid chemistry and efficacy; narrowed candidates for mouse and NHP studies.
- Ran all hands-on experiments on a two-person team and programmed Hamilton liquid handlers for characterization.
- Used Design of Experiments to map parameter interactions, and scaled batches from 1 mL to 1 L for NHP studies.
- Joined as a contractor and promoted to Senior Research Associate; presented to teams across Lilly sites.

Takeda Pharmaceuticals: Oncology Drug Discovery – Cambridge, MA — Jul '19 – Dec '19

Analytical Chemistry Co-op

- Ran purity, stability, and solubility assays on drug candidates; presented an in vivo ELISA study to 50+ scientists.

Publications and Presentations

Hengen, A.C.; **Kolon, H.**; Ackun-Farmmer, M.A. "Biomaterial-mediated immunomodulation of regulatory T cells in cancer." *Biomaterials Science* 2026, 14, 4567–4593. DOI: 10.1039/d6bm00653a

Kolon, H.; Mabrouk, P.A. "How Participation in a Research Experiences for Undergraduates Program (REU) Transforms Chemistry and Biochemistry Majors' Understanding of Research and Scientific Inquiry." *Journal of Chemical Education* 2022, 99, 2493. DOI: 10.1021/acs.jchemed.1c00999

Kolon, H.; Zhang, C. "Lipid Nanoparticles for Direct-to-CNS Delivery." *Internal Eli Lilly Poster Session*, Dec 2023.

Zhang, C.; **Kolon, H.** "A High-Throughput Screening Platform for Lipid Nanoparticle Formulation." *Internal Eli Lilly Poster Session*, Dec 2022.